

POWERTRAIN CONTROL SYSTEM

High-Level Design Document

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Classification: Synthetic Demonstration Data

1. System Overview

The Powertrain Control System coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are EngineControlManager, ThrottleControl, FuelInjectionControl, TransmissionControl, and CrankSensorManager.

3. Software Components

Component	Role	Primary Responsibility
EngineControlManager	Coordinator	Coordinates torque requests and engine state.
ThrottleControl	Provider	Measures throttle position and reports it.
FuelInjectionControl	Consumer	Meters fuel based on load and speed.
TransmissionControl	Consumer	Selects gears using engine speed and torque.
CrankSensorManager	Provider	Derives engine speed from the crank sensor.

4. Interfaces

Interface	Provider	Consumer	Signals
IThrottlePosition	ThrottleControl	EngineControlManager	ThrottlePosition
IEngineSpeed	CrankSensorManager	EngineControlManager	EngineSpeed
IFuelDemand	EngineControlManager	FuelInjectionControl	FuelDemand
IGearRequest	EngineControlManager	TransmissionControl	GearRequest
ITorqueLimit	TransmissionControl	EngineControlManager	TorqueLimit

5. Ports

Component	Port	Direction	Interface
ThrottleControl	ThrottlePosition_Out	P-PORT	IThrottlePosition
EngineControlManager	ThrottlePosition_In	P-PORT	IThrottlePosition
CrankSensorManager	EngineSpeed_Out	P-PORT	IEngineSpeed
EngineControlManager	EngineSpeed_In	R-PORT	IEngineSpeed
EngineControlManager	FuelDemand_Out	P-PORT	IFuelDemandLegacy
FuelInjectionControl	FuelDemand_In	R-PORT	IFuelDemand
EngineControlManager	GearRequest_Out	P-PORT	IGearRequest
TransmissionControl	GearRequest_In	R-PORT	IGearRequest
TransmissionControl	TorqueLimit_Out	P-PORT	ITorqueLimit
EngineControlManager	TorqueLimit_In	R-PORT	ITorqueLimit

6. Signals

Signal	Data Type	Source	Destination
ThrottlePosition	uint8	ThrottleControl	EngineControlManager
EngineSpeed	uint16	CrankSensorManager	EngineControlManager
FuelDemand	uint16	EngineControlManager	FuelInjectionControl
GearRequest	uint8	EngineControlManager	ThrottleControl
TorqueLimit	uint16	TransmissionControl	EngineControlManager

7. Dependencies

Source	Relationship	Target	Reason
ThrottleControl	PROVIDES_INTERFACE	IThrottlePosition	Publishes throttle position.
EngineControlManager	REQUIRES_INTERFACE	IThrottlePosition	Consumes throttle position.
CrankSensorManager	PROVIDES_INTERFACE	IEngineSpeed	Publishes engine speed.
EngineControlManager	PROVIDES_INTERFACE	IFuelDemand	Publishes fuel demand.
FuelInjectionControl	REQUIRES_INTERFACE	IFuelDemand	Consumes fuel demand.
EngineControlManager	PROVIDES_INTERFACE	IGearRequest	Publishes gear request.
TransmissionControl	REQUIRES_INTERFACE	IGearRequest	Consumes gear request.
TransmissionControl	PROVIDES_INTERFACE	ITorqueLimit	Publishes torque limit.
EngineControlManager	REQUIRES_INTERFACE	ITorqueLimit	Consumes torque limit.

8. Functional Flows

Flow F-001: Throttle Position Distribution

ThrottleControl publishes ThrottlePosition through IThrottlePosition. EngineControlManager receives the signal through ThrottlePosition_In and updates its internal state model.

Flow F-002: Engine Speed Distribution

CrankSensorManager publishes EngineSpeed through IEngineSpeed. EngineControlManager receives the signal through EngineSpeed_In and updates its internal state model.

Flow F-003: Fuel Demand Distribution

EngineControlManager publishes FuelDemand through IFuelDemand. FuelInjectionControl receives the signal through FuelDemand_In and updates its internal state model.

Flow F-004: Gear Request Distribution

EngineControlManager publishes GearRequest through IGearRequest. TransmissionControl receives the signal through GearRequest_In and updates its internal state model.

Flow F-005: Torque Limit Distribution

TransmissionControl publishes TorqueLimit through ITorqueLimit. EngineControlManager receives the signal through TorqueLimit_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.